Jiseung Jeong

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RESEARCH INTERESTS

Efficient and Interpretable On-Device Foundation Models
Currently focusing on Vision-Language-Action (VLA) systems

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) Ulsan, Korea
B.S. in Computer Science and Engineering Mar 2021 — Present
GPA: 3.95/4.3 (Major: 4.01/4.3)

RESEARCH AND WORK EXPERIENCE

Ubiquitous AI Lab @ UNIST Ulsan, Korea
Undergraduate Research Intern Apr 2026 — Present

- Reduced mean wall-clock time of a frozen Vision-Language-Action (VLA) policy by 17.9% on a real SO-101 robot (60.5 s → 49.7 s) and 8.3% on LIBERO while maintaining task success, by starting execution from partial instructions via a vision-language focus map and an action readiness gate.

Ubiquitous AI Lab @ UNIST Ulsan, Korea
Undergraduate Research Intern Mar 2025 — Jul 2025

- Researched on-device chain-of-thought (CoT) reasoning, focusing on CoT profiling.

LABIS (Startup team, PSD file translation service) Ulsan, Korea
Software Engineer Oct 2024 — Jul 2025

- Developed text-layer and style extraction for Photoshop Documents; implemented parallel translation processing with Django Celery and minimized data transferred to the server.
- Optimized post-translation rendering via dynamic wait-time detection, reducing loading delay by approximately 50%.

Republic of Korea Army Cheorwon, Korea
Military service Mar 2023 — Sep 2024

TEACHING EXPERIENCE

HD Hyundai Heavy Industries AI Novatus Academia, UNIST Ulsan, Korea
Teaching Assistant, Project-Based Learning (Advisor: Prof. Taesik Gong) Sep 2026 — Present

- Mentoring an industry team building an LLM agent that flags anomalies in large-engine shop-test reports; designed a rule-based extraction + LLM judgment pipeline and demonstrated a prototype on an on-premise open-weight model.

LG Electronics Living DX Course (LDC), UNIST Graduate School of AI Ulsan, Korea
Teaching Assistant, Project-Based Learning (Advisor: Prof. Taesik Gong) Jul 2025 — Aug 2025

- Mentored an industry team analyzing customer usage logs of LG Styler's dehumidification course, guiding preprocessing, usage-pattern analysis, and product-improvement proposals.

PUBLICATIONS

P: Preprint

[**P.1**] "Premover: Fast Vision-Language-Action Control by Acting Before Instructions Are Complete"

Joonha Park, **Jiseung Jeong**, Taesik Gong
Preprint, 2026.

HONORS AND AWARDS

Manifold Sponsor Track Award (Hackathon) Apr 2022

- Awarded for developing an AI-based service at the hackathon hosted by Passfinder Inc.
- Built a web-based application with an interactive interface for configuring CNN model layers and hyperparameters.

PROJECTS

QuantShift (1st / 21 students) Spring 2026
Course project for CSE402 (UNIST). Showed that self-speculative decoding preserves acceptance under 4/8-bit quantization, and that its speedup is bounded by the AR baseline—so NF4, whose single-token decode is already fast, cannot be accelerated by tuning the exit depth.

Shadow Boxing Tracking App with 3 Devices (1st / 8 teams, final presentation) Spring 2026
Course project for CSE465 (UNIST). Recognized and logged boxing motions by fusing IMU data from an earbud, phone, and smartwatch.

YOLOkemon App Spring 2026
Course project for CSE465 (UNIST). A YOLO-based Pokédex object-collection app using a post-training-quantized (PTQ) model.

Fused Dequantization for Quantized LLM Attention (1st / 13 teams) Fall 2025
Course project for CSE412 (UNIST). Designed CUDA kernels that fuse block-wise 4-bit dequantization (NF4, MXFP4, NVFP4) into the QKV projection, achieving 20–25% lower attention latency for NF4/MXFP4 on an RTX 3090.

Lane Detection with Depth Completion — BTS (Brain-to-Society Program, UNIST) 2022
Sensor fusion of stereo camera and LiDAR for depth completion and lane detection, implemented in PyTorch.

RELEVANT COURSES

UNIST *(all completed with an A+ grade)*

- AI Programming II
- System Programming
- Algorithm
- Mathematical Foundations of Machine Learning
- Computer Network
- Parallel Computing
- Mobile Computing
- Natural Language Processing

SKILLS

- **Certification:** COS PRO Level 1 (C++ Data Structures & Algorithms), 2022.07
- **Programming languages:** Python, C/C++